

PAVAN KUMAR SADASHIV

Full-Stack Systems & Applied AI Engineer | Founder & Managing Director

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Platform: hrlpavan.github.io/hrl-international-website- | Charter: github.com/hrlpavan/engineering-compensation-blueprint

EXECUTIVE SUMMARY

Full-stack systems engineer and applied AI architect with a track record of building production software across low-level graphics computing, combinatorial optimization algorithms, and enterprise AI systems. Proven expertise in developing hardware-accelerated C++/Metal/CUDA plugins for Hollywood-grade post-production suites (DaVinci Resolve), architecting NP-hard constraint solvers with zero-dependency native Node.js backends, and directing enterprise multi-agent AI pipelines. Founder of HRL International Private Limited with demonstrated competencies in corporate grant financing (Startup India / Karnataka ELEVATE), distributed stream processing (JPMorgan Chase certified), and enterprise cybersecurity governance (Deloitte certified).

TECHNICAL SKILLS MATRIX

Low-Level & GPU Computing:	C++17/20, Apple Metal Shading Language (MSL), NVIDIA CUDA Kernel Programming, OpenFX API, ACEScc Color Science, 12-Bit DPX Sensor Processing, ONNX Runtime.
Algorithms & Optimization:	Combinatorial Discrete Optimization, Constraint Satisfaction Problem (CSP) Backtracking, Dynamic K-Means Clustering, Bayesian Classifiers, Sliding Window Rate Limiting.
Backend & Systems:	Native Node.js (node:http, node:fs), SQLite 3 (better-sqlite3, ACID Transactions), REST API Design, Microservices, Zero-Framework Server Engineering.
AI/ML & Multimodal:	PyTorch, Monocular Depth Estimation (Depth Anything), Autonomous Agentic AI (Google Antigravity SDK), ElevenLabs Neural Voice DSP, Cloud Composer (Airflow), BigQuery.
Frontend & Visual:	JavaScript (ES6+), HTML5 2D Canvas API, Tailwind CSS, Apple Liquid Glass UI Design, Responsive Web Applications, jsPDF, SheetJS.
DevOps & Infrastructure:	Git, GitHub Actions (CI/CD), Automated Deployments, Linux, macOS Application Bundling & DMG Packaging.

FLAGSHIP ENGINEERING PROJECTS

1. AI Cinematic Haze — OpenFX & Fusion Plugin for DaVinci Resolve

Specialization: Low-Level C++, Apple Metal & CUDA Compute, Computer Vision AI

- Architected a high-performance OpenFX visual computing plugin integrated directly into DaVinci Resolve and Blackmagic Fusion.
- Engineered dual hardware-accelerated GPU backends using Apple Metal Shading Language for Apple Silicon and CUDA kernels for NVIDIA GPUs.
- Integrated monocular neural depth estimation (Depth Anything via ONNX Runtime) to calculate physically accurate volumetric optical atmospheric scattering.
- Enforced strict Hollywood ACEScc color science preserving 12-bit uncompressed DPX sensor dynamic range without luminance clipping.

2. InvigiMatrix — College Examination & Invigilation Management Portal

Specialization: Combinatorial Optimization (CSP), Zero-Dependency Full-Stack, Machine Learning |

github.com/hrlpavan/exam-invigilation-dsa-app

- Implemented an NP-hard Constraint Satisfaction Problem (CSP) Backtracking solver allocating faculty duties, rooms, and timetables with zero collisions.
- Built an AI/ML Smart Batch Planner with K-Means (k=3) & Bayesian classifier for anti-cheating student seat dispersion.
- Engineered a high-throughput, zero-framework native Node.js REST server backed by SQLite with strict ACID-compliant transactions.

3. OmniTransform AI — Intelligent Process Automation Platform

Specialization: Multimodal AI Pipelines, Acoustic Signal Processing, Enterprise IPA | github.com/hrlpavan/omnitransform-ai-resources

- Architected an enterprise IPA platform integrating multimodal document parsing and phonetic entity extraction.
- Developed a neural speech pipeline leveraging ElevenLabs Multilingual v2 Voice AI paired with Chladni resonance acoustic physics visualizers.

LEADERSHIP & ENTREPRENEURSHIP

Founder & Managing Director — HRL International Private Limited (2026 – Present)

- Founded enterprise deep-tech software venture; authored Startup India Seed Fund & Karnataka ELEVATE grant dossiers.
- Directed media distribution infrastructures achieving 2.5M+ organic impressions across digital creator channels.

EDUCATION & CERTIFICATIONS

Bachelor of Engineering (B.E.) in Computer Science & Engineering (AI & ML)

Sahyadri College of Engineering & Management (SCEM), Mangaluru, Karnataka, India

- **JPMorgan Chase & Co.** — Distributed Streaming & Software Engineering Simulation (Kafka, REST Controller Architecture)
- **Deloitte Australia** — Cybersecurity Defense & Enterprise Threat Modeling Simulation (Perimeter Defense, Incident Response)
- **Blackmagic Design** — DaVinci Resolve Color Science & Visual Computing Integration